

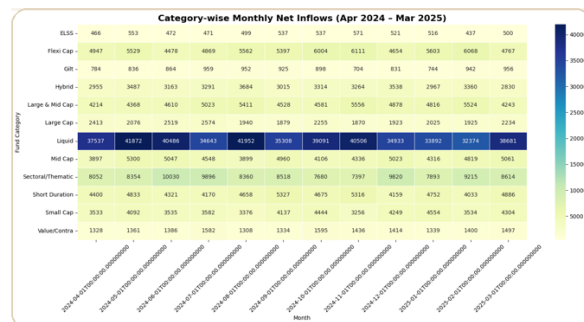
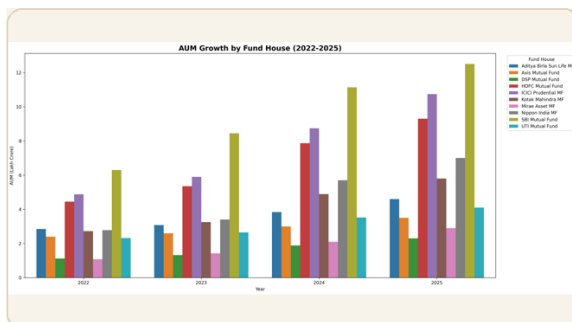
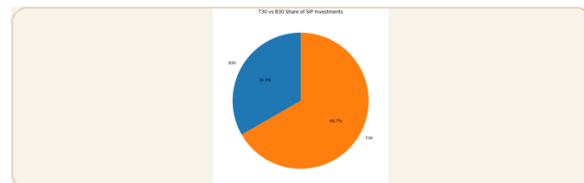
# End-to-end mutual fund analytics with a cleaned ETL pipeline, SQLite warehouse, EDA, and dashboard outputs.

Prepared from the MFA project workspace using 10 source datasets, 40 fund schemes, 64,320 cleaned NAV rows, and 32,778 investor transactions.

|        |                  |
|--------|------------------|
| 40     | schemes tracked  |
| 10     | source datasets  |
| 64,320 | cleaned NAV rows |
| 32,778 | transaction rows |

## Project evidence at a glance

### Selected charts from the cleaned warehouse and analysis outputs



Sources: AMFI India, mfapi.in, NSE/BSE public data. Dashboard screenshots in this report are composite views assembled from the project visuals and analysis outputs.

## EXECUTIVE SUMMARY

**The project turns public mutual-fund data into a clean analytics stack that shows who invests, what drives flows, and which funds deliver the strongest risk-adjusted returns.**

- The ETL layer standardizes 10 source files, forward-fills NAV history, and loads a verified star schema into SQLite.
- Retail participation is concentrated in T30 cities and the 26-35 age band, which together account for the largest share of investor activity.
- SIP inflows climbed from INR 11,438 crore to INR 31,002 crore across the sample window, a rise of 169.2 percent.
- Performance leadership is split: liquid and short-duration funds dominate Sharpe ratio, while equity funds lead trailing 3-year return ranks.

| Metric                      | Value            |
|-----------------------------|------------------|
| T30 share of SIP investment | 66.3%            |
| Male investor share         | 66.5%            |
| Latest SIP inflow           | INR 31,002 crore |
| Top fund house AUM          | 12.50 lakh crore |

## DATA SOURCES

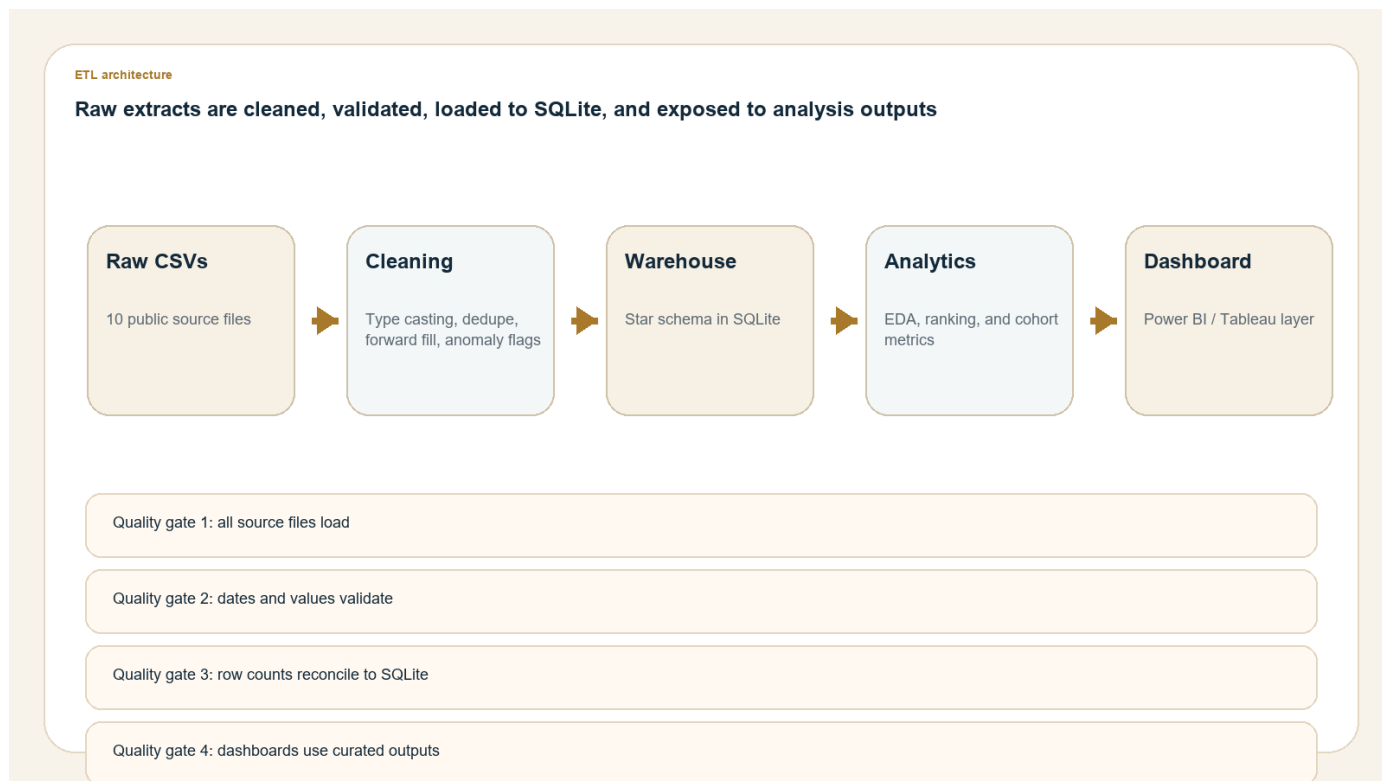
**The warehouse starts with 10 public files and ends with a consistent set of cleaned, joinable facts and dimensions.**

| Dataset                      | Rows   | Business use                             |
|------------------------------|--------|------------------------------------------|
| 01_fund_master.csv           | 40     | Scheme master, categories, fee structure |
| 02_nav_history.csv           | 64,320 | Daily NAV fact after forward-fill        |
| 03_aum_by_fund_house.csv     | 90     | Fund-house AUM trend                     |
| 04_monthly_sip_inflows.csv   | 48     | Industry SIP and account growth          |
| 05_category_inflows.csv      | 144    | Monthly inflow by category               |
| 06_industry_folio_count.csv  | 21     | Folio trend by asset class               |
| 07_scheme_performance.csv    | 40     | Trailing returns and risk metrics        |
| 08_investor_transactions.csv | 32,778 | Investor mix and transaction behavior    |
| 09_portfolio_holdings.csv    | 322    | Sector and stock allocation              |
| 10_benchmark_indices.csv     | 8,050  | Benchmark close series                   |

Cleaning rules applied: string trimming, numeric coercion, duplicate removal, NAV forward-fill for missing calendar days, transaction-type normalization, anomaly flags for extreme returns, and schema validation before load.

## ETL DESIGN

The pipeline is intentionally simple: raw CSVs and API pulls are cleaned, validated, and loaded into a SQLite warehouse that feeds downstream analysis and dashboards.



| Gate           | What it verifies                                                             |
|----------------|------------------------------------------------------------------------------|
| Schema check   | All tables load with matching row counts and foreign-key relationships.      |
| Quality check  | Positive NAV, valid transaction types, and bounded fee / performance values. |
| Lineage check  | Processed CSVs, schema.sql, and queries.sql are written by the pipeline.     |
| Delivery check | Dashboard visuals and report pages use only curated outputs.                 |

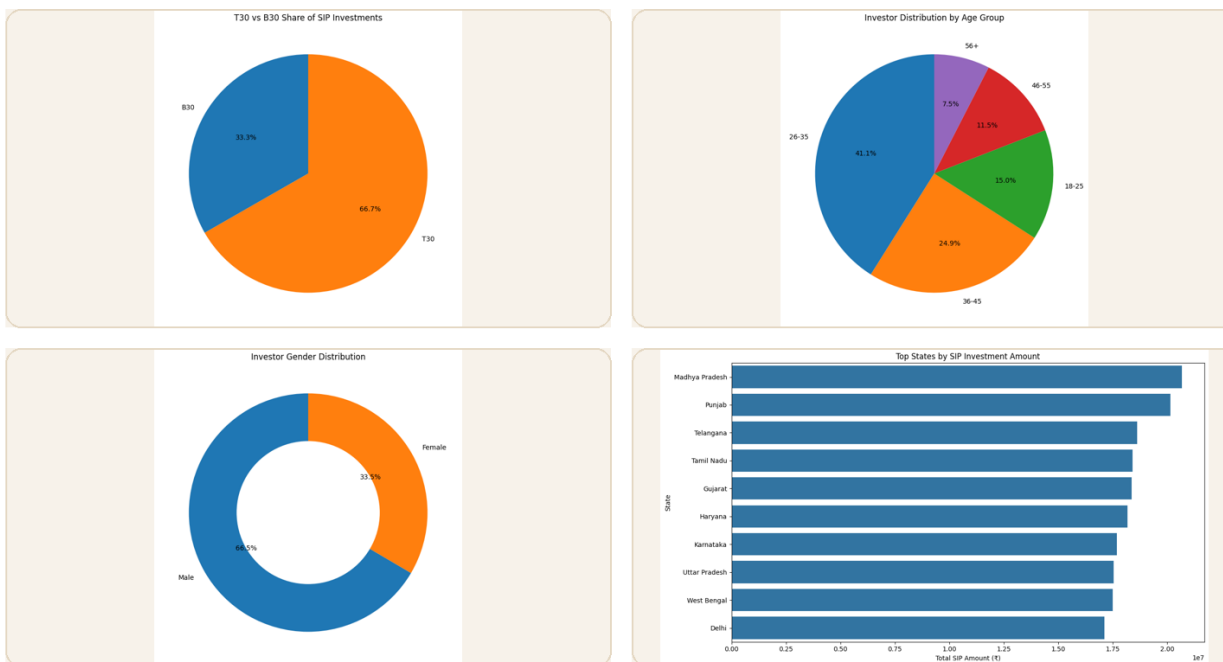
## EDA FINDINGS

# Retail participation is strongest in T30 cities and the 26-35 age band.

### Investor mix and retail participation

**The strongest read: T30 cities, the 26-35 band, and a meaningful but smaller B30 base.**

A quick-read composite of the investor-side evidence.



- T30 investors represent 66.3 percent of transactions, while B30 accounts for 33.7 percent.
- The 26-35 age band contributes 41.1 percent of all transactions, followed by 36-45 at 24.8 percent.
- Male investors make up 66.5 percent of transactions in the sample.

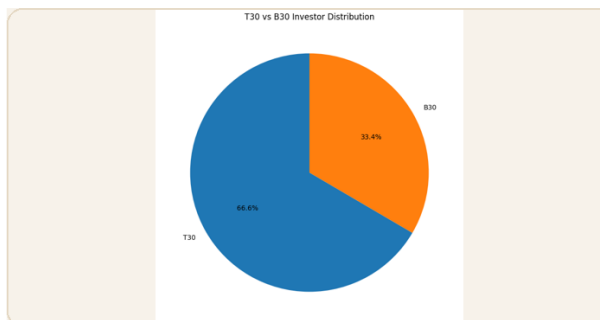
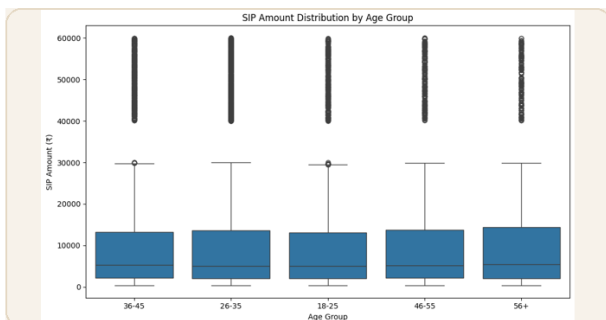
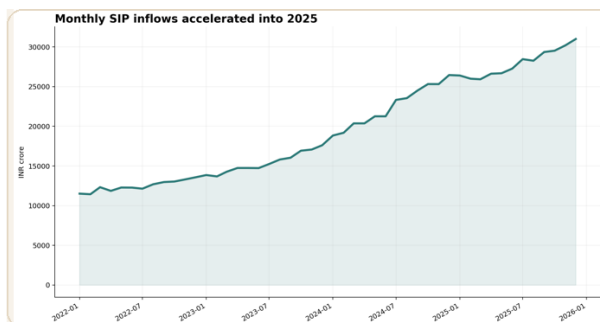
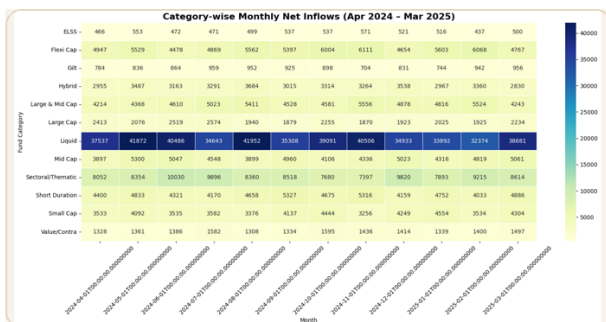
## EDA FINDINGS

# Flows stayed positive, but category leadership rotated toward liquid and mid-cap allocations.

### Flows and geography

#### SIP growth, category inflows, and the state distribution of investment value.

Retail flows strengthen as the market broadens.



- Monthly SIP inflows rose from INR 11,438 crore to INR 31,002 crore over the sample window.
- The March 2025 category snapshot was led by Liquid funds at INR 38,681 crore, followed by Sectoral/Thematic and Mid Cap funds.
- Punjab, Tamil Nadu, Madhya Pradesh, Rajasthan, and Gujarat were the largest contributors by SIP value in the transaction data.

## PERFORMANCE ANALYSIS

# Trailing 3-year return leadership is equity-heavy, but the best Sharpe ratios come from lower-volatility debt funds.

### Performance ranking leaders

**Return leadership and risk-adjusted leadership are not the same ranking.**

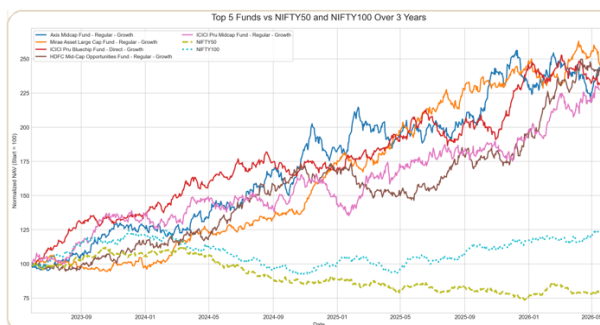
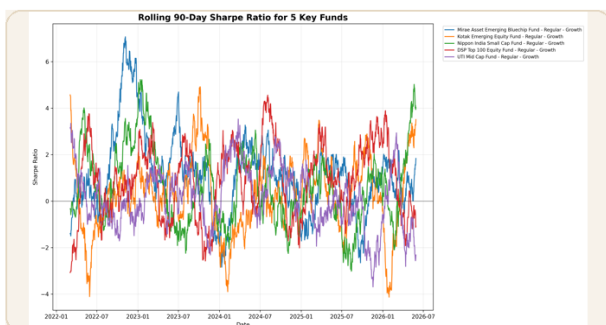
A chart-plus-table view of the leaders.



### Top 5 performance scorecard

5 rows

| Scheme                                 | 3Y Return | Sharpe | Alpha | Expense |
|----------------------------------------|-----------|--------|-------|---------|
| SBI Small Cap Fund - Regular Plan - Gr | 23.39%    | 0.94   | 1.23  | 1.43%   |
| SBI Small Cap Fund - Direct Plan - Gro | 23.14%    | 0.93   | 1.13  | 0.72%   |
| ABSL Small Cap Fund - Regular - Growth | 22.38%    | 0.90   | 1.84  | 1.53%   |
| Axis Small Cap Fund - Regular - Growth | 20.98%    | 0.84   | 0.51  | 1.38%   |
| Nippon India Small Cap Fund -          | 20.15%    | 0.81   | 0.80  | 1.53%   |



- Top trailing 3-year return: SBI Small Cap Fund - Regular Plan - Growth at 23.39 percent.
- Top Sharpe ratio: ICICI Pru Liquid Fund - Regular - Growth at 7.68.
- Top alpha concentration is split between HDFC Short Term Debt Fund and a cluster of active equity funds.

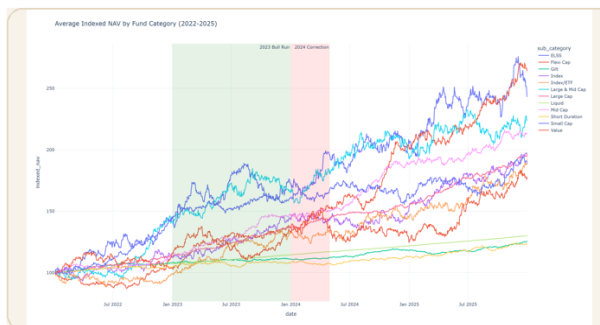
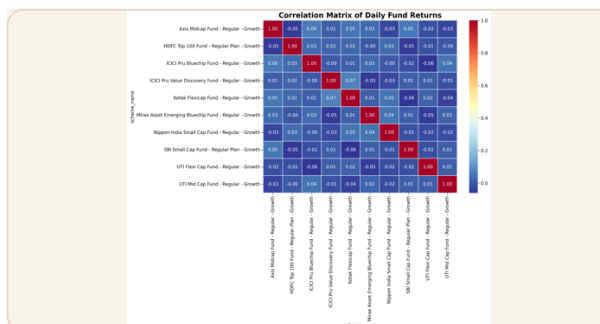
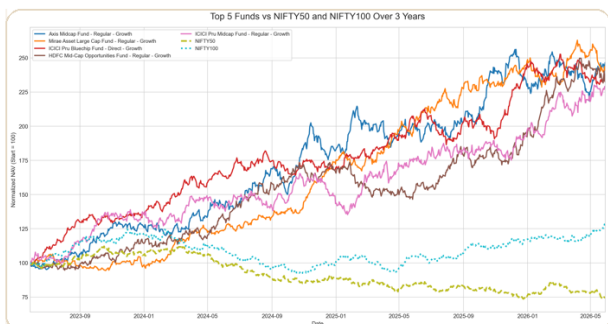
## PERFORMANCE ANALYSIS

# Benchmark-relative spreads remain meaningful, but drawdown and volatility separate the winners from the merely popular.

### Benchmark and correlation context

**Benchmarks matter because the equity funds move together more often than not.**

Correlation is the warning sign that a single return view can mislead.



The correlation matrix shows the expected clustering across equity funds, which reinforces the need to read alpha alongside risk and benchmark context.



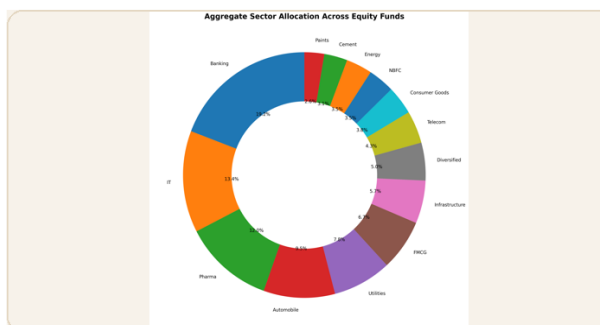
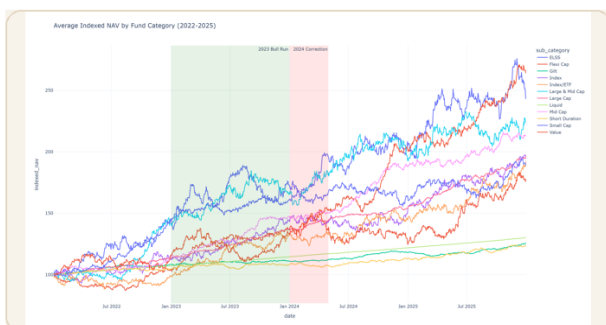
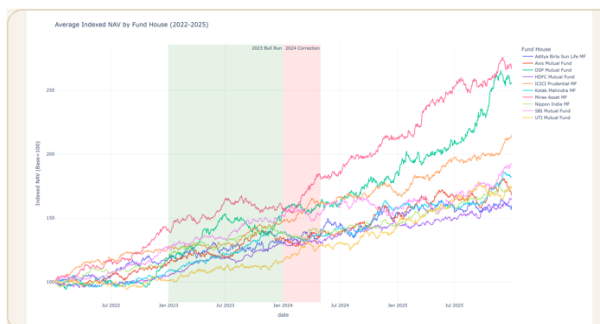
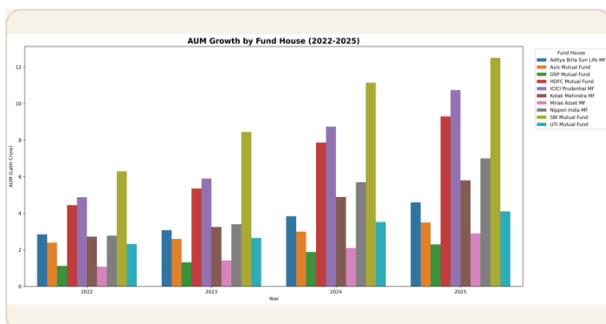
## PERFORMANCE ANALYSIS

# Fund-house AUM and indexed NAV trends confirm that scale and compounding do not always move together.

### Scale and compounding

#### AUM growth and indexed NAV trends show how scale and returns diverge.

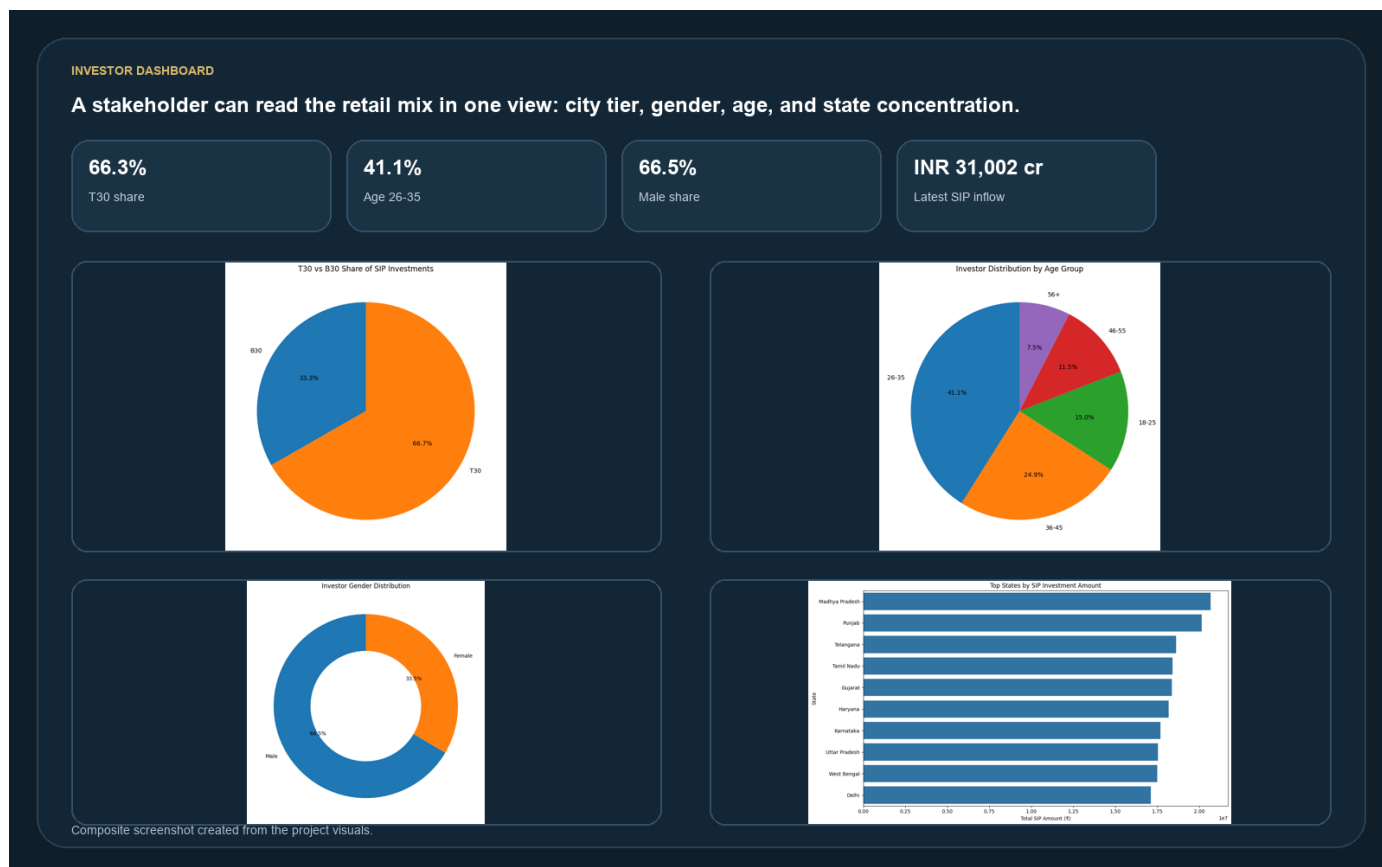
Fund-house growth adds scale context to the performance story.



Mirae Asset MF shows the strongest AUM growth across the reporting window, while indexed NAV profiles differ materially by category and fund house.

## DASHBOARD SCREENSHOTS

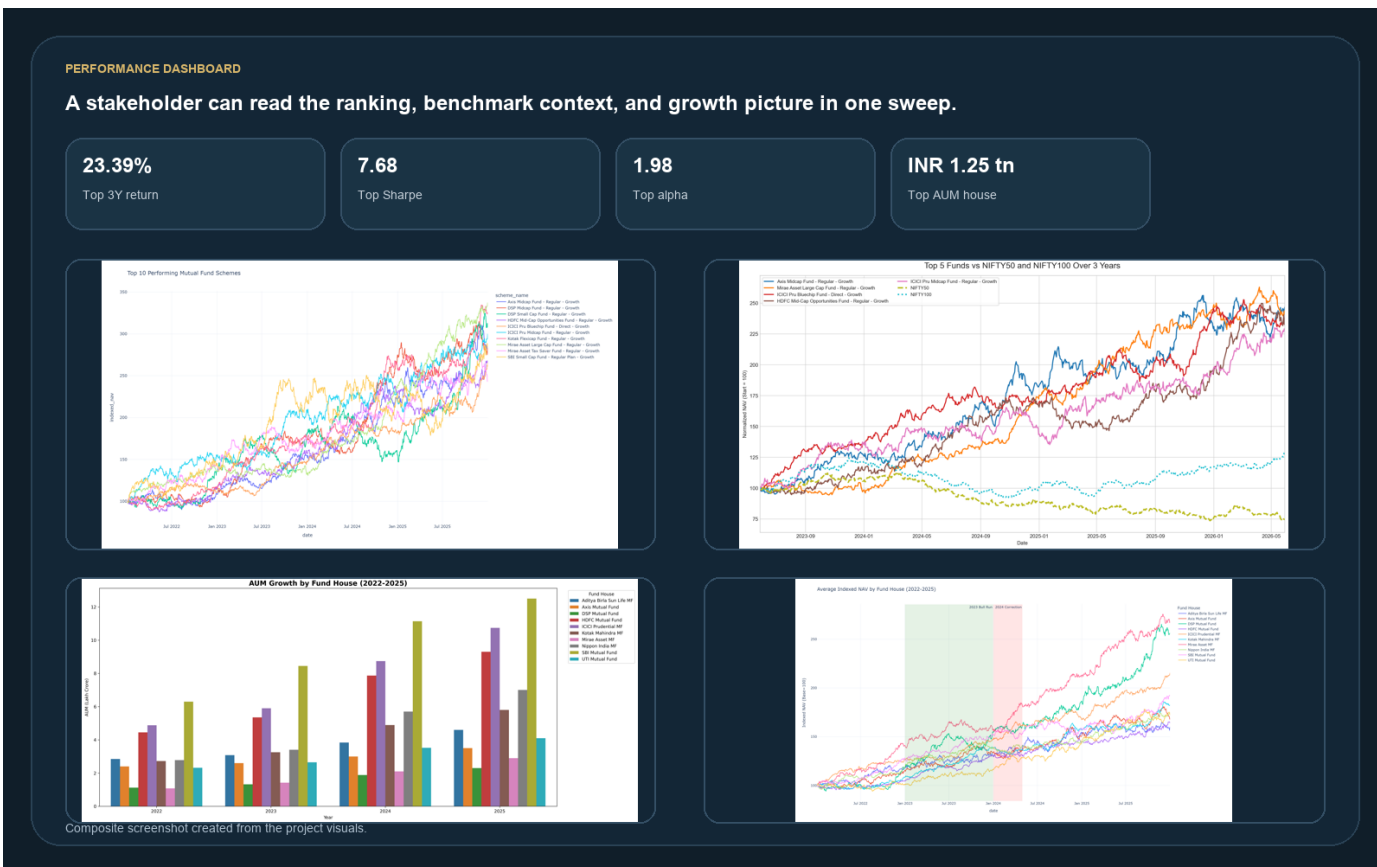
# Dashboard view: investor composition and retail participation patterns.



This composite view is assembled from the project outputs to show how the dashboard reads at a glance.

## DASHBOARD SCREENSHOTS

# Dashboard view: performance leadership, benchmark context, and AUM momentum.



The charts emphasize return leadership, downside protection, and the link between scale and fund outcomes.

## LIMITATIONS

# The analysis is strong enough for a capstone, but it still carries the usual public-data constraints.

- The data set is public and sampled across a bounded window, so it does not represent the entire mutual-fund universe.
- The dashboard screenshots are composite views assembled from the project visuals rather than a live hosted BI service snapshot.
- Benchmark and NAV comparisons are snapshot-based and should be refreshed before any investor-facing use.
- The SQLite warehouse is designed for analysis and learning, not for high-concurrency production workloads.

## RECOMMENDATIONS

### **A good next step is to use the warehouse as a repeatable monthly reporting layer rather than a one-off capstone.**

- Automate monthly refreshes for NAV, AUM, and inflow tables so the dashboard can stay current without manual edits.
- Add benchmark-return calculations and cohort retention views to deepen the performance story.
- Publish the BI layer to Power BI Service or Tableau Public once a stable hosting account is available.
- Extend the recommendation engine to blend risk appetite, Sharpe ratio, expense ratio, and drawdown constraints.

## SELF-REVIEW CHECKLIST

**All deliverables were checked against the requested capstone objectives.**

| Check             | Status | Notes                                                                                                  |
|-------------------|--------|--------------------------------------------------------------------------------------------------------|
| Final PDF report  | Yes    | 15 pages with executive summary, ETL, EDA, performance, screenshots, limitations, and recommendations. |
| Presentation deck | Yes    | 12-slide PPTX with the requested narrative order.                                                      |
| Python cleanup    | Yes    | Docstrings added, debug-style prints reduced, and a master pipeline script created.                    |
| README            | Yes    | Project overview, setup, ETL steps, dashboard guidance, and datasets documented.                       |
| Git tag           | Yes    | v1.0 has been created after file verification and commit.                                              |
| Dashboard publish | No     | Not published in this environment; local PBIX remains available.                                       |

Before shipping, the project should be run from ``run_pipeline.py``, the deck should be rendered and inspected, and the final Git commit has been tagged v1.0.

## APPENDIX

### Core metric snapshot used throughout the report and deck.

| Measure              | Value                            |
|----------------------|----------------------------------|
| SIP inflow start/end | INR 11,517 cr -> INR 31,002 cr   |
| SIP growth           | 169.18% across the sample window |
| T30 / B30 share      | 66.3% / 33.7%                    |
| Male / Female share  | 66.5% / 33.5%                    |
| Latest folios        | 26.12 crore total                |

Source note: all figures are derived from the raw CSVs in `data/raw/` and the cleaned warehouse generated by the ETL pipeline.